



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Professional Skill Certificate

Semester 1 2025/2026

GLRT0060 Safety and Health Officer Introductory Course

Section 1

Individual Tasks

Trainer: Mohd Saidin bin Misnan

Date: 29 November 2025

Student Name	Lam Yoke Yu
Matric Number	A23CS0233
IC Number	040211-08-0642
Email Address	lam.yu@graduate.utm.my
Year and Program	3/SECPH
Programme Name	Bachelor in Computer Science (Data Engineering) with Honours
School/Faculty Name	Faculty of Computing

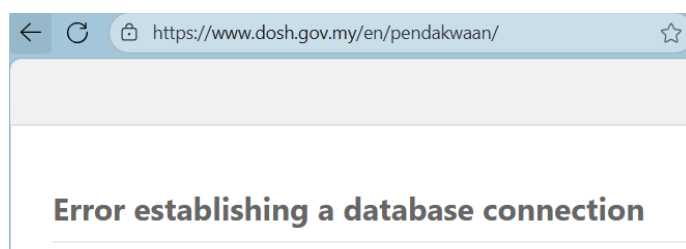
Task 1: Safety (in your profession)

Safety means practice of designing, developing, and maintaining computer systems, software, and digital infrastructures in a way that prevent harm, protects users, and ensures reliable and secure operation. For both employees and technology, safety measures include ergonomics, mental health support, physical safety, cybersecurity, and electrical safety.

Task 2: Accident Prosecution Cases

Please review three (3) accident prosecution cases related to your discipline/profession from the latest sources or as shown below.

* The DOSH system is not working, only get to find one before the system is down.



No	Company	Date	Description
1	The Phoenix Press Sdn. Bhd.	12 July 2024	Failing at all times to maintain all safety equipment and machinery. Source: https://www.dosh.gov.my/en/pendakwaan/
2	Alibaba Cloud services	10 September 2024	Lithium battery explosion at a Singapore data centre. Source: https://www.thestar.com.my/tech/tech-news/2024/09/11/fire-at-data-centre-disrupts-alibaba-cloud-services-affecting-lazadaandtiktok Although no injuries were reported in this incident, it still represents a significant safety hazard.
3	The Forestry Department of Peninsular Malaysia	16 September 2016	The ICT room was destroyed in a fire on Jalan Sultan Salahuddin. No casualties were reported, as the department's office was closed for Malaysia Day. Source: https://www.malaymail.com/news/malaysia/2016/09/16/forestry-dept-fire-over-rm20m-computer-equipment-destroyed/1207455 Although no injuries were reported in this incident, it still represents a significant safety hazard.

Task 3: Safety and Health Officer (SHO)

There are 9 duties of SHO listed in the regulation.

1. to advise the employer or any person in charge of a place of work on the measures to be taken in the interest of the safety and health of the persons employed in the place of work

2. to inspect the place of work to determine whether any machinery, plant, equipment, substance, appliances or process or any description of manual labour used in the place of work, is of such nature liable to cause bodily injury to any person working in the place of work
3. to investigate any accident, near-miss accident, dangerous occurrence, occupational poisoning, or occupational disease which has happened in the place of work
4. to assist the employer or the safety and health committee, if any, pursuant to regulation 11 of the Occupational Safety and Health (Safety and Health Committee) Regulations 1996 [P.U.(A) 616/96] in organising and implementing occupational safety and health programme at the place of work
5. to become secretary to the safety and health committee, if any, as specified under subregulation 6(2) of the Occupational Safety and Health (Safety and Health Committee) Regulations 1996, and perform all functions of a secretary as specified in that Regulations
6. to assist the safety and health committee in any inspection of the place of work for the purposes of checking the effectiveness and efficacy of any measures taken in compliance with the Act or any regulations made under the Act
7. to collect, analyse and maintain statistics on any accident, dangerous occurrence occupational poisoning and occupational disease which have occurred at the place of work
8. to assist any officer in carrying made by the employer or any person in charge of the place of work on any matters pertaining to safety and health of the place of work
9. A safety and health officer shall, before the tenth of a preceding month, submit a report pertaining to his activities to the employer.

Task 4: OSHA 1994

Please list the duty of employer based on OSHA1994 and Court case related to

1. It shall be the duty of every employer to ensure, so far as is practicable, the safety, health, and welfare at work of all his employees.
2. the provision and maintenance of plant and systems of work that are, so far as is practicable, safe and without risks to health
3. the making of arrangements for ensuring, so far as is practicable, safety and absence of risks to health in connection with the use or operation, handling, storage and transport of plant and substances
4. the provision of such information, instruction, training and supervision as is necessary to ensure, so far as is practicable, the safety and health at work of his employees
5. so far as is practicable, as regards any place of work under the control of the employer, the maintenance of it in a condition that is safe and without risks to health and the provision and maintenance of the means of access to and egress from it that are safe and without such risks

Task 5: OSHA 1994

Please list the duty of employee based on OSHA1994 and Court case related to.

1. to take reasonable care for the safety and health of himself and of other persons who may be affected by his acts or omissions at work
2. to co-operate with his employer or any other person in the discharge of any duty or requirement imposed on the employer or that other person by this Act or any regulation made thereunder
3. to wear or use at all times any protective equipment or clothing provided by the employer for the purpose of preventing risks to his safety and health
4. to comply with any instruction or measure on occupational safety and health instituted by his employer or any other person by or under this Act or any regulation made thereunder
5. No employer shall levy or permit to be levied on any employee of his any charge in respect of anything done or provided in pursuance of this Act or any regulation made thereunder.

Task 6: OSHA 1994

Please list FIVE regulations under OSHA 1994:

<https://www.dosh.gov.my/index.php/legislation/eregulations>

1. Occupational Safety and Health (Employer's Safety and Health General Policy Statements) (Exception) Regulations 1995
2. Occupational Safety and Health (Control of Industrial Major Accident Hazard) Regulations 1996
3. Occupational Safety and Health (Safety and Health Committee) Regulations 1996
4. Occupational Safety and Health (Classification, Packaging and Labeling of Hazardous Chemicals) Regulations 1997
5. Occupational Safety and Health (Safety and Health Officer) Regulations 1997

Task 7: OSH related Guidelines

Please list FIVE OSH related guidelines related to your study discipline/profession

<https://www.dosh.gov.my/index.php/legislation/guidelines>

1. The employer shall ensure that employees and their safety and health representatives are consulted, informed and trained on all aspects of OSH, including emergency arrangements, associated with their work
2. The employer shall have overall responsibility for the protection of employees' safety and health and provide leadership for OSH activities in the organisation
3. The organization should maintain up-to-date documentation that is sufficient to ensure that its OSHMS can be adequately understood and effectively and efficiently implemented.

4. The organization shall estimate and evaluate the level of risks associated with the identified hazards. The results of these assessments shall be considered for decision making and for determining control.
5. The organization should actively assess potential accident and emergency response needs, develop plans, procedures and processes to cope with them, test its planned responses, and seek to improve the effectiveness of its responses.

Task 8: Accident Causation

Please review three (3) Fatal Accident cases related to your discipline/profession from the latest sources or as shown below.

<https://www.dosh.gov.my/index.php/fatal-accident-case-1>

Based on my research, there are no recorded cases related to my discipline in Malaysia; therefore, cases from other countries are discussed instead.

No	Location	Date	Description
1	QTS Data Center at Georgia, USA	25 September 2025	Anthony J. Riera Azuaje , a worker at a QTS data-centre construction site in Fayetteville (Georgia) was electrocuted while performing work duties. He was transported to hospital but died. According to a lawsuit filed by his family, the subcontractor responsible for electrical work (Allison-Smith Company) allegedly violated electrical safety standards: they reportedly installed wrong fittings, failed to properly ground cables, and left a metal conduit energized — which caused the fatal electrocution when the worker (employed by another contractor) made incidental contact with it while on a scissor lift. Source: https://www.datacenterdynamics.com/en/news/lawsuit-launched-after-death-of-construction-worker-at-data-center-site-in-fayetteville-georgia/
2	Turner Construction at Hyperscale Data Center, Medina County, Texas, USA	6 October 2025	Maurice McGowan was fatally struck by an articulating truck at the construction site of a new large-scale data centre (the “Cinco Project”). The worker was pronounced dead on site. The site is a major data-centre campus under development by Rowan Digital Infrastructure, with Turner Construction as general contractor. Source: https://www.enr.com/articles/61581-worker-identified-in-fatal-turner-construction-data-center-accident
3	Cyber 1 Data Center at Jakarta, Indonesia	2 December 2021	A fire broke out at the Cyber 1 data centre in Jakarta. Two technicians inside the server room died from smoke inhalation after servers “exploded.” Source: https://www.datacenterdynamics.com/en/news/two-

			killed-in-fire-at-cyber-1-data-center-in-jakarta-indonesia/
--	--	--	---

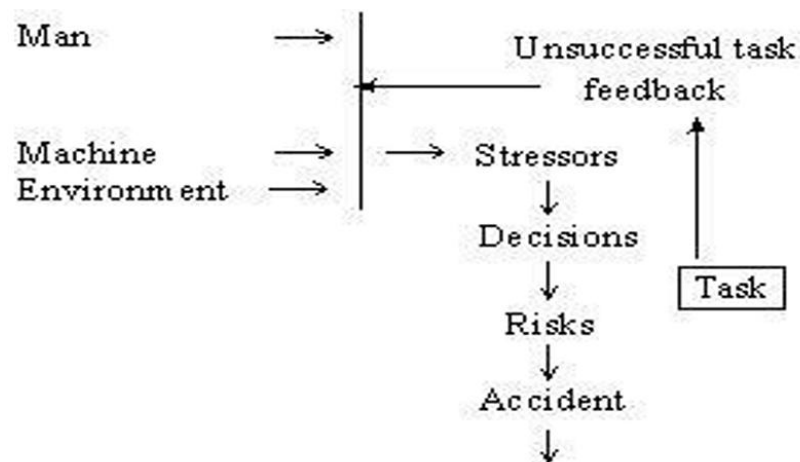
Task 9: Accident Theory

Please review some of the accident theories from the latest sources (internet etc)

1. Systems Model Theory (Bob Firenze)

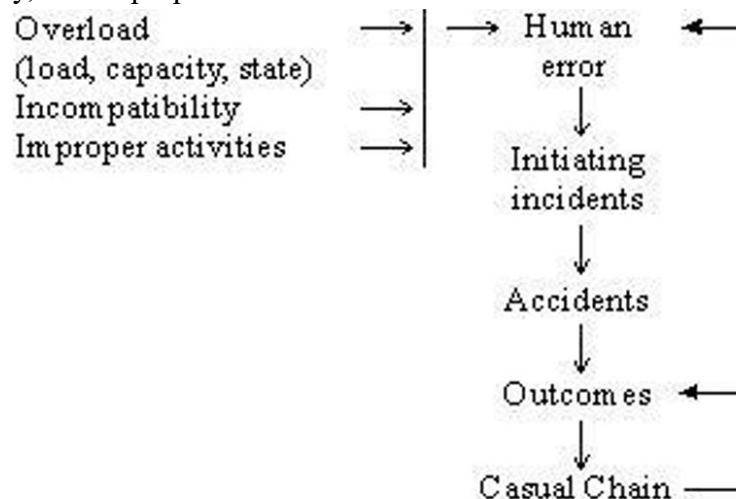
Interaction between man, machine, and environment leads to an accident if the information available to the important element of the system is inadequate.

If the risk is high and the decisions based on information are illogical and unsound, an accident occurs resulting into incompleteness of the task. Bob Firenze's system model is shown below:



2. Ferrell's Human Factors Theory

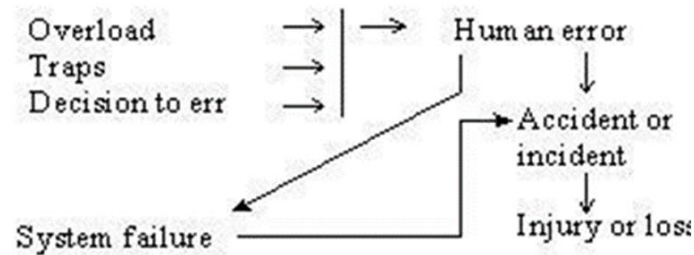
This theory states that accidents are the result of a casual chain, one or more of the causes being human error, which is in turn caused by three situations - overload, incompatibility, and improper activities.



3. Petersen's Accident-Incident Causation Theory

This theory adapts Ferrell's human factors of overload and states that causes of accident/incident are human error and/or system failure. Human error is due to overload, traps and decision to error. Human error may directly cause accident or may cause

system failure which may cause accident resulting in injury or loss as shown in the diagram.



Source: https://cosstraining.org/wp-content/uploads/2023/11/M06_AccidentCausationTheories-PDF.pdf

Task 10: Unsafe Act vs Unsafe Condition

Please review some unsafe act vs unsafe condition at workplace (internet etc)

1. Not wearing personal protective equipment (PPE) (Unsafe Act)
2. Operating machinery at unsafe speeds (Unsafe Act)
3. Blocking emergency exits (Unsafe Act)
4. Poor lighting in work areas (Unsafe Condition)
5. Exposed electrical wiring (Unsafe Condition)

Working at Height

Types of job working at height

1. Construction
2. Bridge Inspection and Repair
3. Wind Turbine Maintenance

Task 11: Safety Policy

Please identify UTM safety policy (from establish sources)

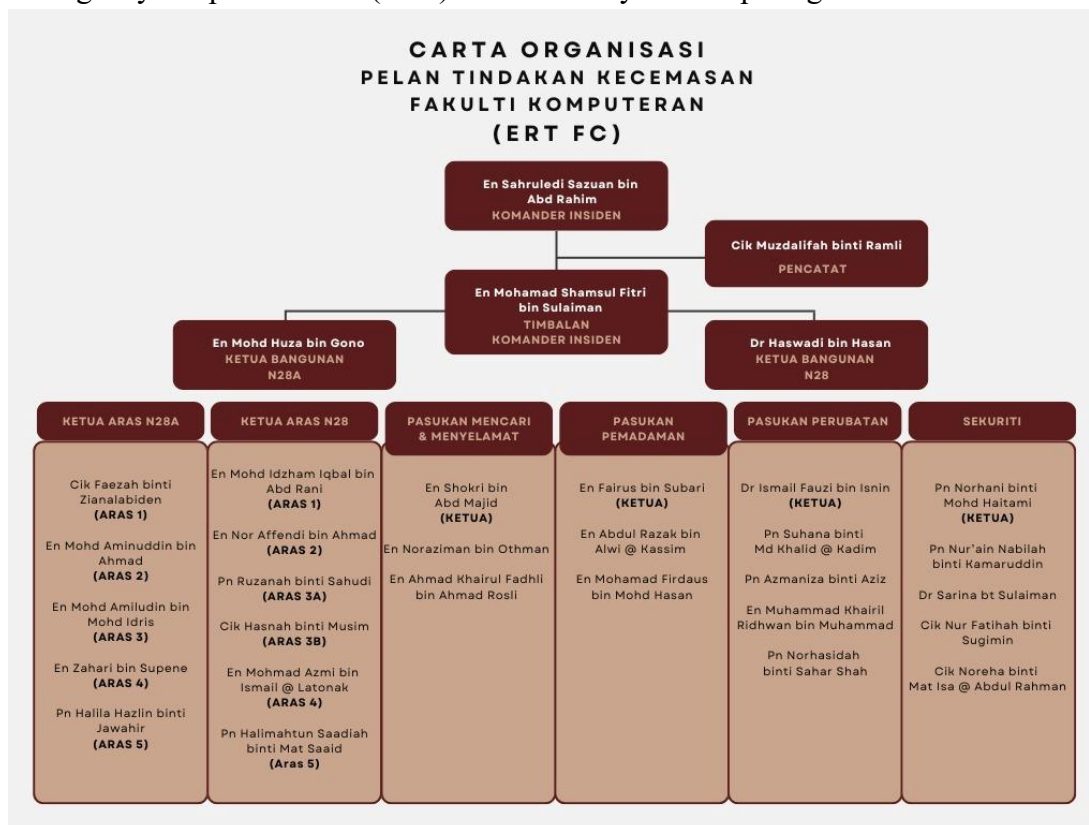
1. UTM is committed towards establishment and implementation of an effective Occupational Safety and Health Management System to continually improve our performances, minimisation of hazards and prevention of accident and injuries in the premises for all our activities during routine and non-routine operations.
2. Signed by Prof. Dr. Mohd Shafry bin Mohd Rahim (Vice-Chancellor of UTM) and date 9 January 2025

Task 12: Safety Management Plan

1. Safety and Health Committee of the Faculty of Computing



2. Emergency Response Team (ERT) of the Faculty of Computing



3. UTM Emergency Contact No.



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SENARAI TALIAN KECEMASAN

TALIAN KECEMASAN UMUM



POLIS



BOMBA



HOSPITAL



PERTAHANAN AWAM (JPAM)



PUSAT RACUN NEGARA.....04-653 6999

999



TALIAN KECEMASAN DALAMAN



BAHAGIAN KESELAMATAN UTM..... 07-533 3013 / 07-553 0002



PUSAT KESIHATAN UTM.....07-553 0999



PEJABAT OSHE UTM..... 07-553 1886

TALIAN KECEMASAN SEKITAR UTM

BALAI POLIS	BALAI POLIS SKUDAI.....	07-560 5750
	BALAI POLIS TAMAN UNIVERSITI.....	07-520 3129
	BALAI POLIS KANGKAR PULAI.....	07-527 1255
	BALAI POLIS SENAI.....	07-599 1222
	BALAI POLIS MUTIARA RINI.....	07-557 1122
BALAI BOMBA & PENYELAMAT	BBP TAMAN UNIVERSITI, SKUDAI.....	07-520 4144
	BBP JOHOR BAHRU.....	07-224 3444
	BBP PEKAN NANAS.....	07-699 4444
	BBP KULAI.....	07-663 4444
KLINIK / HOSPITAL	KLINIK KESIHATAN TAMAN UNIVERSITI.....	07-521 6800
	HOSPITAL SULTANAH AMINAH.....	07-225 7000
	HOSPITAL SULTAN ISMAIL.....	07-356 5000
	HOSPITAL KULAI.....	07-662 3333

TALIAN JABATAN BERKAITAN

JABATAN KESELAMATAN & KESIHATAN PEKERJAAN JOHOR.. 07-207 9900
JABATAN ALAM SEKITAR NEGERI JOHOR..... 07-550 0522



<https://dvcdev.utm.my/oshe/>



Menginovasi Penyelesaian

www.utm.my



Task 13: Spot the Hazard

Please identify at least seven hazards at the workplace as shown in the picture

